

Assignment 1 – NLP

Assignment Objective

Develop a Natural Language Processing (NLP) prototype that combines a **lexicon-based assessment method** with **LLM-based processing**, using self-developed assessment resources and models available through Ollama. The assignment should cover the full development cycle, from conceptual design to implementation, testing, and critical evaluation of a **locally deployable solution**.

1. Product Purpose

- Define the problem the prototype is intended to solve.
- Explain the relevance of the proposed solution.
- Identify the potential users, domains, or practical applications of the prototype.

2. Model or Method

- Explain the principles of the **lexicon-based assessment method** adopted in the prototype.
- Justify the choice of lexicon, whether self-developed or obtained from third-party sources.
- Explain the role of the **LLM as the processing component** of the solution.
- Describe how the lexicon-based assessment and the LLM-based processing interact within the overall system.
- If external repositories, datasets, or open-source resources are used, cite them properly.

3. Solution Design for Local Deployment

- Present the overall structure and workflow of the solution.
- Describe the system architecture required for local deployment.
- Identify the libraries, packages, models, and other dependencies necessary for implementation.
- Explain how Ollama is integrated into the workflow, including the selected model or models and their specific function in the prototype.
- Indicate the hardware and software requirements needed to run the solution locally.

4. Implementation Code

- Develop and document the source code of the solution.
- Explain the most relevant portions of the code.
- Follow good programming practices, ensuring readability, modularity, and appropriate commenting.
- Organize the implementation so that it can be executed and tested in a local environment.

5. Product Functionality Testing

- Demonstrate the execution of the prototype and present the results obtained.
- Explain the results clearly and analytically.
- Discuss the effectiveness of the lexicon-based assessment and the LLM-based processing.
- Suggest possible improvements, refinements, or future extensions.

6. Development Experience

- Report the main challenges encountered during development.
- Explain how these challenges were addressed.
- Reflect on the knowledge and skills acquired throughout the project.

7. Market Potential

- Indicate whether the product could have value in a specific area, sector, or market.
- Present realistic scenarios in which the solution could be applied.
- Discuss possible users, beneficiaries, or future applications.

Submission

The assignment must be submitted in **PDF report format**. The report should include all relevant components of the project, such as explanatory comments, code excerpts, architecture description, outputs, results, and analytical discussion, following the structure defined in this guide.

Evaluation Criteria

The assignment will be evaluated according to the following criteria:

- Creativity and innovation in the developed solution.
- Clarity in the presentation of the problem and proposed solution.
- Coherence in the description of the lexicon-based assessment method and the LLM-based processing.
- Quality and completeness of the documentation.
- Quality of implementation, testing, and analysis of results.
- Feasibility and consistency of the local deployment design.